

Statistics from samples and Limit theorems

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Subsection 1

Statistics from *iid* samples

Where have we seen *iid* samples?

- Bernoulli trials
- Monte carlo simulations
- Computing histograms

Bernoulli trials

- Experiment and an event of interest A
 - ▶ Occurance of an Event A is considered success
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Goal: Try to estimate $P(A) = P(X_i = 1)$

- Useful in finding prevalence of a disease in a population etc.

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iid samples: $X_1, X_2, \dots, X_n$

- $X_i = 1$ if A occurs in the i -th trial, and $X_i = 0$ otherwise
- Estimate $P(A) = P(X_i = 1)$ (similar to Bernoulli trial)

Computing histograms

- n data points of some variable of interest
 - ▶ $x_1, x_2, \dots, x_n$
- Bin: $[a, b]$
 - ▶ n_b : number of x_i that fall inside $[a, b]$

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- Event $A = (a < X < b)$

Histogram count

- Data points: *iid* samples $X_1, X_2, \dots, X_n \sim f_X(x)$
- Estimate $P(a < X < b) \approx n_b/n$

iid samples hold information on distribution

- What is common to all 3 of the previous scenarios?
 - ▶ Given: *iid* samples
 - ▶ Goal: get some partial information about the distribution
 - ★ Procedures to gather the information needed

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- *iid* samples of an unknown or partially known distribution form the input for statistical procedures
 - ▶ Data: modelled as observations from *iid* repetitions of an experiment
 - ▶ Example: Iris data
 - ★ Data from every iris is considered to be *iid* observations from the distribution of the 4 lengths

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Analysis

- How to decide if the statistical procedure is “good”?
- How many samples are needed for a “goodness” guarantee?

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- **Goal**

- ▶ Find p from *iid* samples

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- Sampling 1

- ▶ 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1

- Sampling 2

- ▶ 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1

- Sampling 3

- ▶ 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

- and so on. . . .

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- and so on. . . .

- **Important**

- ▶ p is the same for all samplings
- ▶ However, samples do not remain the same
- ▶ Each sample in each sampling is an observation of a random variable

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- **Requirement on statistical procedure**

- ▶ In spite of variations in samples, provide p with some guarantee

What is a typical statistical problem?

- **Model for Samples:** $X_1, X_2, \dots, X_n \sim \text{iid } X$
Handwritten notes:
- X : PMF $p_X(x)$
- Each X_i : PMF $p_X(x)$
- X_i are independent
- **Given "data":** $x_1, x_2, \dots, x_n$ from one sampling instance
- Distribution of X is partially known or unknown
 - ▶ What is partially known? Know distribution but parameters unknown
 - ▶ Example: Bernoulli(p) with p unknown, Normal(μ, σ^2) with μ and σ unknown
- **Goal:** Procedures to find information about the distribution of X

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- **Goal:** Procedures to find information about the distribution of X
- **What information?**
 - ▶ What is the mean of X ? What is the variance of X ?
 - ▶ What is $P(X > t)$? What is $P(a < X < b)$?
 - ▶ What is the distribution of X ? What is the size of T_X ?

Subsection 2

Empirical distribution and descriptive statistics

Empirical distribution

Definition (Empirical distribution)

Let $X_1, X_2, \dots, X_n \sim X$ be iid samples. Let $\#(X_i = t)$ denote the number of times t occurs in the samples. The empirical distribution is the discrete distribution with PMF

$$p(t) = \frac{\#(X_i = t)}{n}.$$

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- 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1

Empirical $\frac{8}{20}$ $\frac{12}{20}$
Distribution $\{0, 1\}$

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- 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1
 - ▶ $p(0) = 7/20, p(1) = 13/20$

Empirical Distribution $\left\{ \begin{matrix} 7/20 & 13/20 \\ 0 & 1 \end{matrix} \right\}$

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$\{0, 1, 2, 3\}$

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- 1, 2, 0, 3, 0, 0, 1, 2, 0, 1, 3, 2, 1, 1, 0, 3, 0, 2, 2, 1
▶ $p(0) = 6/20$, $p(1) = 6/20$, $p(2) = 5/20$, $p(3) = 3/20$

$\left\{ \begin{matrix} 6/20 & 6/20 & 5/20 & 3/20 \\ 0, 1, 2, 3 \end{matrix} \right\}$

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- **Descriptive statistics:** Properties of empirical distribution
 - ▶ Mean of the distribution
 - ▶ Variance of the distribution
 - ▶ Probability of an event
- As number of samples increases, the properties of empirical distribution *should* become close to that of the original distribution

Sample mean

Definition (Sample mean)

Let $X_1, X_2, \dots, X_n$ be iid samples. The sample mean, denoted $\bar{X}$, is defined to be the random variable

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

- Given a sampling $x_1, \dots, x_n$, the value taken by the sample mean $\bar{X}$ is $\bar{x} = (x_1 + \dots + x_n)/n$. Often, $\bar{X}$ and $\bar{x}$ are both called sample mean.

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 - Value taken by $\bar{X}$: 12/20

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 - Value taken by $\bar{X}$: $12/20$
- 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1
 - Value taken by $\bar{X}$: $13/20$
- 1, 2, 0, 3, 0, 0, 1, 2, 0, 1, 3, 2, 1, 1, 0, 3, 0, 2, 2, 1
 - Sample mean: $25/20$

Illustration 1: Bernoulli(0.5) samples

$$X_1, \dots, X_n \sim \text{iid } \left\{ \begin{matrix} 1/2 & 1/2 \\ 0 & 1 \end{matrix} \right\}$$

- $n = 5$

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$$X_1, \dots, X_n \sim \text{iid } \{0, 1\}^{1/2 \ 1/2}$$

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 - ▶ Samples: 0, 1, 1, 1, 0; Sample mean: $3/5$
- $n = 20$
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 - ▶ 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1; $13/20$
- $n = 200$

Illustration 1: Bernoulli(0.5) samples

$$X_1, \dots, X_n \sim \text{iid } \{0, 1\}^{1/2 \ 1/2}$$

- $n = 5$
 - ▶ Samples: 0, 0, 1, 1, 1; Sample mean: $3/5$
 - ▶ Samples: 1, 1, 1, 0, 1; Sample mean: $4/5$
 - ▶ Samples: 0, 1, 1, 1, 0; Sample mean: $3/5$
- $n = 20$
 - ▶ 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1; $12/20$
 - ▶ 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1; $13/20$
 - ▶ 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1; $13/20$
- $n = 200$
 - ▶ Sampling 1: $95/200$, Sampling 2: $102/200$, Sampling 3: $98/200$

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 - ▶ Samples: 0, 0, 1, 1, 1; Sample mean: $3/5$
 - ▶ Samples: 1, 1, 1, 0, 1; Sample mean: $4/5$
 - ▶ Samples: 0, 1, 1, 1, 0; Sample mean: $3/5$
- $n = 20$
 - ▶ 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1; $12/20$
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 - ▶ 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1; $13/20$
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Illustration 1: Bernoulli(0.5) samples

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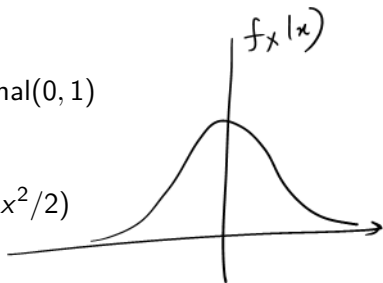
Distribution mean = $\frac{1}{2}$

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 - ▶ Samples: 1, 1, 1, 0, 1; Sample mean: $4/5$
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 - ▶ 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1; $12/20$
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 - ▶ 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1; $13/20$
- $n = 200$
 - ▶ Sampling 1: $95/200$, Sampling 2: $102/200$, Sampling 3: $98/200$
- $n = 1000$
 - ▶ Sampling 1: $495/\overset{10}{1000}$, Sampling 2: $490/\overset{10}{1000}$, Sampling 3: $504/\overset{10}{1000}$

Illustration 2: Normal(0, 1) samples

$$X_1, \dots, X_n \sim \text{iid Normal}(0, 1)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$$



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- $n = 5$

- ▶ Samples: 2.17, 0.10, -0.75, -1.05, -1.72; Sample mean: -0.25

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- $n = 200$
 - ▶ Sampling 1: -0.01, Sampling 2: 0.11, Sampling 3: -0.12

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- $n = 1000$

Illustration 2: Normal(0, 1) samples

$$X_1, \dots, X_n \sim \text{iid Normal}(0, 1)$$

↑
Distribution mean

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- $n = 20$
 - ▶ Sampling 1: 0.08, Sampling 2: -0.24, Sampling 3: 0.41
- $n = 200$
 - ▶ Sampling 1: -0.01, Sampling 2: 0.11, Sampling 3: -0.12
- $n = 1000$
 - ▶ Sampling 1: 0.04, Sampling 2: -0.04, Sampling 3: -0.02

Expected value and variance of sample mean

Theorem

Let $X_1, X_2, \dots, X_n$ be iid samples whose distribution has a finite mean μ and variance σ^2 . The sample mean $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ has expected value and variance given by

$$E[\bar{X}] = \mu, \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

$$\begin{aligned} E[\bar{X}] &= \frac{E[X_1] + E[X_2] + \dots + E[X_n]}{n} = \frac{n\mu}{n} = \mu \\ E[\bar{X}^2] &= \frac{E[X_1^2] + \dots + E[X_n^2] + 2E[X_1X_2] + \dots + 2E[X_{n-1}X_n]}{n^2} \\ &= \frac{nE[X^2] + n(n-1)\mu^2}{n^2} = \frac{n\sigma^2 + n\mu^2 + n^2\mu^2 - n\mu^2}{n^2} \\ &= \frac{\sigma^2}{n} + \mu^2 \\ \text{Var}(\bar{X}) &= \frac{\sigma^2}{n} \end{aligned}$$

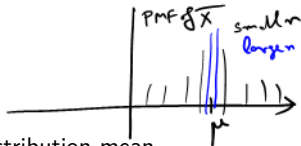
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- Expected value of sample mean equals the expected value or mean of the distribution
 - Mean of distribution: constant real number and not random
 - Sample mean: random variable with mean equal to distribution mean
- Variance of sample mean decreases with n
 - As n increases. . .
 - variance of sample mean tends to zero
 - the spread of sample mean will decrease
 - sample mean will take values close to the distribution mean



Sample variance

Definition (Sample variance)

Let $X_1, X_2, \dots, X_n$ be iid samples. The sample variance, denoted S^2 , is defined to be the random variable

$$S^2 = \frac{(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n - 1},$$

where $\bar{X}$ is the sample mean.

- Given a sampling $x_1, \dots, x_n$, the value taken by the sample variance S^2 is $s^2 = ((x_1 - \bar{x})^2 + \dots + (x_n - \bar{x})^2)/(n - 1)$. Often, S^2 and s^2 are both called sample variance.

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- Why $n - 1$ in the denominator instead of n ?
 - ▶ Some books use n (this causes confusion)
 - ▶ Expected value of sample variance is simple in this case

Expected value of sample variance

Theorem

Let $X_1, X_2, \dots, X_n$ be iid samples whose distribution has a finite variance σ^2 . The sample variance $S^2 = \frac{(X_1 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n-1}$ has expected value given by

$$E[S^2] = \sigma^2.$$

Expected value of sample variance

Theorem

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$$E[S^2] = \sigma^2.$$

- Expected value of sample variance equals the variance of the distribution
 - ▶ Variance of distribution: constant real number and not random
 - ▶ Sample variance: random variable with mean equal to distribution variance
- Values of sample variance, on average, give the variance of distribution
 - ▶ Variance of sample variance will decrease with number of samples (in most cases)
 - ▶ As n increases, sample variance takes values close to distribution variance

Illustration

- Bernoulli($1/2$), mean = 0.5, variance = 0.25
 - ▶ Sample variance values: $n = 20$
 - ★ 0.26, 0.26, 0.26, 0.25, 0.26
 - ▶ Sample variance values: $n = 200$
 - ★ 0.2500, 0.2487, 0.2496, 0.2456, 0.2476
 - ▶ Sample variance values: $n = 1000$
 - ★ 0.2498, 0.2490, 0.2499, 0.2501, 0.2502

Illustration

- Bernoulli($1/2$), mean = 0.5, variance = 0.25

- ▶ Sample variance values: $n = 20$
 - ★ 0.26, 0.26, 0.26, 0.25, 0.26
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 - ★ 0.2500, 0.2487, 0.2496, 0.2456, 0.2476
- ▶ Sample variance values: $n = 1000$
 - ★ 0.2498, 0.2490, 0.2499, 0.2501, 0.2502

- Normal(0,1), mean = 0, variance = 1

- ▶ Sample variance values: $n = 20$
 - ★ 0.89, 0.57, 1.19, 1.01, 1.41
- ▶ Sample variance values: $n = 200$
 - ★ 0.93, 1.07, 0.85, 0.83, 1.09
- ▶ Sample variance values: $n = 1000$
 - ★ 1.0268, 0.9535, 0.9781, 0.9766, 0.9831

Sample proportion

$$X_1, X_2, \dots, X_n \overset{iid}{\sim} X$$

- *iid* samples from the distribution of X
- Let A be an event defined using X
 - ▶ Example: $A = (X > t)$, $A = (a < X < b)$ etc

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The sample proportion of A , denoted $S(A)$, is defined as

$$S(A) = \frac{\#(X_i \text{ for which } A \text{ is true})}{n}.$$

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$$S(A) = \frac{\#(X_i \text{ for which } A \text{ is true})}{n}.$$

- Samples: 0, 1, 1, 1, 0
 - ▶ $S(X = 1) = 3/5$
- Samples: -0.2, 1.1, 0.3, -1.2, 0.7
 - ▶ $S(X \leq 0) = 2/5$, $S(0 < X < 1) = 2/5$, $S(X > 1) = 1/5$

Expected value and variance of sample proportion

Theorem

Let $X_1, X_2, \dots, X_n$ be iid samples from the distribution of X . Let A be an event defined using X and let $P(A)$ be the probability of A . The sample proportion of A , denoted $S(A)$, has expected value and variance given by

$$E[S(A)] = P(A), \text{Var}(S(A)) = \frac{P(A)(1 - P(A))}{n}.$$

Proof

- Convert samples into Bernoulli($P(A)$) samples $Y_1, \dots, Y_n$
 - ▶ $Y_i = 1$ if A is true for X_i , and $Y_i = 0$ otherwise
- $S(A)$ is the sample mean of $Y_1, \dots, Y_n \sim \text{iid Bernoulli}(A)$

mean = $P(A)$
variance = $P(A)(1 - P(A))$

Expected value and variance of sample proportion

Theorem

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 - ▶ $Y_i = 1$ if A is true for X_i , and $Y_i = 0$ otherwise
- $S(A)$ is the sample mean of $Y_1, \dots, Y_n$
- As n increases, values of $S(A)$ will be close to $P(A)$
 - ▶ Mean of $S(A)$ equals $P(A)$
 - ▶ Variance of $S(A)$ tends to 0

Illustration

$$X_1, \dots, X_n \sim \text{Normal}(0, 1)$$

• $P(X \leq -1) = 0.159$ ← from distribution $\int_{-\infty}^{-1} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 0.159 \dots$

- ▶ Sample proportion values: $n = 20$

★ 0.15, 0.20, 0.15, 0.15, 0.15 (5 samplings)

- ▶ Sample proportion values: $n = 200$

★ 0.170, 0.140, 0.150, 0.155, 0.165 (5 samplings)

- ▶ Sample proportion values: $n = 1000$

★ 0.160, 0.180, 0.162, 0.135, 0.153 (5 samplings)

• $P(-1 < X < 1) = 0.683$

- ▶ Sample proportion values: $n = 20$

★ 0.75, 0.70, 0.55, 0.45, 0.70

- ▶ Sample proportion values: $n = 200$

★ 0.705, 0.690, 0.705, 0.670, 0.720

- ▶ Sample proportion values: $n = 1000$

★ 0.678, 0.678, 0.686, 0.679, 0.681

Where have we seen *iid* samples?

- Bernoulli trials: Sample mean tends to distribution mean
 - ▶ Bernoulli(p) samples
 - ▶ Distribution mean = p
 - ▶ Sample mean = fraction of successes
- Monte carlo simulations
 - ▶ Sample proportion tends to actual probability
- Computing histograms
 - ▶ Sample proportion tends to actual probability

Subsection 3

Illustrations with data

Iris data

- 3 classes of irises: 0, 1, 2
 - ▶ 50 instances of data for each class
 - ▶ Each instance: [sepal length, sepal width, petal length, petal width] (cm)

Iris data

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 - ▶ 50 instances of data for each class
 - ▶ Each instance: [sepal length, sepal width, petal length, petal width] (cm)
- Sepal length of Class 0
 - ▶ Model: *iid* samples according to some unknown distribution
 - ▶ Data: 5.1, 4.9, 4.7, ..., 5.3, 5
 - ▶ Sample mean: 5.006, Sample variance: $0.1242 = 0.3524^2$
 - ▶ $S(\text{Sepal length} > 5) = 22/50$, $S(4.8 < \text{Sepal length} < 5.2) = 20/50$

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- Petal width of Class 3
 - ▶ Model: *iid* samples according to some unknown distribution
 - ▶ Data: 2.5, 1.9, 2.1, $\dots$, 2.3, 1.8
 - ▶ Sample mean: 2.026, Sample variance: $0.0754 = 0.2746^2$
 - ▶ $S(\text{Petal width} > 2) = 23/50$, $S(1.8 < \text{Petal length} < 2.2) = 17/50$

Iris data

- 3 classes of irises: 0, 1, 2
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 - ▶ $S(\text{Petal width} > 2) = 23/50$, $S(1.8 < \text{Petal length} < 2.2) = 17/50$
- Model: how good is the iid samples model?

Taj Mahal air quality

Date	SO2	NO2	PM2.5	PM10
12/4	4	60	77	185
13/4	4	53	65	196
11/4	4	57	72	223
10/4	4	45	68	200
8/4	5	33	52	250
7/4	4	27	67	266
6/4	4	12	60	219
5/4	7	27	70	207
4/4	4	58	100	282
3/4	4	17	55	158
1/4	4	31	37	465
Max	80	80	60	100



- 24-hour average of particles in air, units: micrograms/cubic metre

Taj Mahal air quality sample statistics

- Sample means
 - ▶ SO₂: 4.36, NO₂: 38.18, PM_{2.5}: 65.72, PM₁₀: 241
- Sample ^{variance/}standard deviations
 - ▶ SO₂: 0.9244^2 , NO₂: 17.1803^2 , PM_{2.5}: 15.9002^2 , PM₁₀: 82.6184^2
- S(max exceeded)
 - ▶ SO₂: 0, NO₂: 0, PM_{2.5}: 7/11, PM₁₀: 11/11

Taj Mahal air quality sample statistics

- Sample means
 - ▶ SO₂: 4.36, NO₂: 38.18, PM_{2.5}: 65.72, PM₁₀: 241
- Sample standard deviations
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- S(max exceeded)
 - ▶ SO₂: 0, NO₂: 0, PM_{2.5}: 7/11, PM₁₀: 11/11
- Model
 - ▶ Do you like the iid samples model for this data?
 - ▶ Is the Taj in trouble or not? How do we answer such questions?
 - ▶ How confident are our conclusions when we have looked at just 11 data points?

IPL: Runs scored in Deliveries 0.1, 0.2, 0.3

- Data from 1598 innings
 - ▶ See shared spreadsheet
 - ▶ Download csv from cricsheet.org
- All calculations done using spreadsheets or other computing tools

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- Data from 1598 innings
 - ▶ See shared spreadsheet
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- All calculations done using spreadsheets or other computing tools
- Sample means
 - ▶ 0.1: 0.7347, 0.2: 0.8686, 0.3: 0.9524
- Sample variances
 - ▶ 0.1: 1.4975, 0.2: 1.7961, 0.3: 2.0666
- Sample proportions
 - ▶ S(dot ball) - 0.1: 0.5989, 0.2: 0.5551, 0.3: 0.5338
 - ▶ S(4 or 6) - 0.1: 0.0914, 0.2: 0.1145, 0.3: 0.1302

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- All calculations done using spreadsheets or other computing tools
- Sample means
 - ▶ 0.1: 0.7347, 0.2: 0.8686, 0.3: 0.9524
- Sample variances
 - ▶ 0.1: 1.4975, 0.2: 1.7961, 0.3: 2.0666
- Sample proportions
 - ▶ $S(\text{dot ball})$ - 0.1: 0.5989, 0.2: 0.5551, 0.3: 0.5338
 - ▶ $S(4 \text{ or } 6)$ - 0.1: 0.0914, 0.2: 0.1145, 0.3: 0.1302
- Clear trend from samples
 - ▶ Runs scored increases from 0.1 to 0.3
- Enough data points to be confident in the trend
 - ▶ Agrees with intuition
- Model: Do you like the iid samples model for each delivery?

Subsection 4

Sum of independent random variables

Expected value and variance

Theorem

Let $X_1, X_2, \dots, X_n$ be random variables. Let $S = X_1 + \dots + X_n$ be their sum. Then,

$$E[S] = E[X_1] + \dots + E[X_n].$$

If $X_1, \dots, X_n$ are pairwise uncorrelated, then

$$\text{Var}(S) = \text{Var}(X_1) + \dots + \text{Var}(X_n).$$

- What is pairwise uncorrelated? $E[X_i X_j] = E[X_i]E[X_j]$ for all $i, j, i \neq j$

Handwritten derivation of the variance of a sum of random variables:

$$\begin{aligned} \text{Var}(S) &= E[(S - E[S])^2] \\ &= E[(X_1 - E[X_1]) + (X_2 - E[X_2]) + \dots + (X_n - E[X_n])^2] \\ &= E[(X_1 - E[X_1])^2 + (X_2 - E[X_2])^2 + \dots + (X_n - E[X_n])^2 \\ &\quad + 2(X_1 - E[X_1])(X_2 - E[X_2]) + \dots] \\ &= E[(X_1 - E[X_1])^2] + E[(X_2 - E[X_2])^2] + \dots + E[(X_n - E[X_n])^2] \\ &\quad + 2E[(X_1 - E[X_1])(X_2 - E[X_2])] + \dots \end{aligned}$$

A curved arrow points from the term $E[(X_i - E[X_i])(X_j - E[X_j])]$ to the definition of pairwise uncorrelated: $E[X_i X_j] = E[X_i]E[X_j]$ for all $i, j, i \neq j$.

Expected value and variance

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- Mean of sum is sum of means
- If uncorrelated, variance of sum is sum of variances

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- Mean of sum is sum of means
- If uncorrelated, variance of sum is sum of variances
- If the X_i are independent, they are also uncorrelated
 - ▶ So, above result holds for independent random variables

Extensions of previous result

- Scaling and summing

- ▶ Suppose $S = a_1X_1 + \cdots + a_nX_n$, where a_i are constants
- ▶ $E[S] = a_1E[X_1] + \cdots + a_nE[X_n]$
- ▶ $\text{Var}(S) = a_1^2\text{Var}(X_1) + \cdots + a_n^2\text{Var}(X_n)$, if pairwise uncorrelated

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- iid samples: $X_1, \dots, X_n \sim X$, iid

- ▶ Suppose $S = a_1X_1 + \cdots + a_nX_n$, where a_i are constants
- ▶ $E[S] = (a_1 + \cdots + a_n)E[X]$
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- Sample mean: $X_1, \dots, X_n \sim X$, iid

- ▶ $\bar{X} = (X_1 + \cdots + X_n)/n$, $a_i = 1/n$
- ▶ $E[(\bar{X})] = E[X]$
- ▶ $\text{Var}((\bar{X})) = \text{Var}(X)/n$

Sample mean versus distribution mean

$$X_1, \dots, X_n \sim \text{iid } X$$

- Let $\mu = E[X]$, $\sigma^2 = \text{Var}(X)$

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 - ▶ Expected value: μ , Variance: σ^2/n
 - ▶ Variance (or spread) goes to 0 as n grows

Can we say something more precise about $\bar{X}$ and μ ?

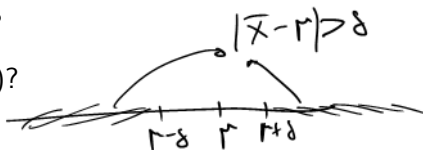
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Can we say something more precise about $\bar{X}$ and μ ?

- What is $P(\bar{X} > \mu + \delta)$?
- What is $P(\bar{X} < \mu - \delta)$?
- What is $P(|\bar{X} - \mu| > \delta)$?



Weak law of large numbers

$$X_1, \dots, X_n \sim \text{iid } X$$

- Let $\mu = E[X]$, $\sigma^2 = \text{Var}(X)$
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Theorem (Weak law of large numbers)

$$P(|\bar{X} - \mu| > \delta) \leq \frac{\sigma^2}{n\delta^2} \rightarrow 0.$$

Chebyshev:

$$P((\bar{X} - \mu)^2 > \delta^2) \leq \frac{E[(\bar{X} - \mu)^2]}{\delta^2} = \frac{\sigma^2}{n\delta^2} \xrightarrow[n \rightarrow \infty]{\text{fixed } \delta} 0$$

Weak law of large numbers

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Theorem (Weak law of large numbers)

$$P(|\bar{X} - \mu| > \delta) \leq \frac{\sigma^2}{n\delta^2} \rightarrow 0.$$

- With probability more than $1 - \frac{\sigma^2}{n\delta^2}$, sample mean lies in $[\mu - \delta, \mu + \delta]$
 - ▶ **What is the meaning of this probability?**

Weak law of large numbers

$$X_1, \dots, X_n \sim \text{iid } X$$

$\bar{X}$: converges to μ
"in probability"

- Let $\mu = E[X]$, $\sigma^2 = \text{Var}(X)$
- Sample mean: $\bar{X} = (X_1 + \dots + X_n)/n$
 - ▶ Expected value: μ , Variance: σ^2/n

Theorem (Weak law of large numbers)

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 - ▶ **What is the meaning of this probability?**
- Chebyshev is usually a very "weak" bound, and we will see sharper bounds soon

Examples: n iid samples

- Bernoulli(p) samples $\mu = p, \sigma^2 = p(1-p)$
 - ▶ With probability more than $1 - \frac{p(1-p)}{n\delta^2}$, sample mean lies in $[p - \delta, p + \delta]$
- Uniform $\{-M, \dots, M\}$ samples $\mu = 0, \sigma^2 = \frac{M(M+1)}{3}$
 - ▶ With probability more than $1 - \frac{M(M+1)}{3n\delta^2}$, sample mean lies in $[-\delta, \delta]$
- Normal($0, \sigma^2$) samples
 - ▶ With probability more than $1 - \frac{\sigma^2}{n\delta^2}$, sample mean lies in $[-\delta, \delta]$
- Uniform $[-A, A]$ samples *Continuous* $\mu = 0, \sigma^2 = \frac{A^2}{3}$
 - ▶ With probability more than $1 - \frac{A^2}{3n\delta^2}$, sample mean lies in $[-\delta, \delta]$

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 - ▶ With probability more than $1 - \frac{p(1-p)}{n\delta^2}$, sample mean lies in $[p - \delta, p + \delta]$
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- Normal($0, \sigma^2$) samples
 - ▶ With probability more than $1 - \frac{\sigma^2}{n\delta^2}$, sample mean lies in $[-\delta, \delta]$
- Uniform $[-A, A]$ samples
 - ▶ With probability more than $1 - \frac{A^2}{3n\delta^2}$, sample mean lies in $[-\delta, \delta]$
- When distribution is known, a precise statement is possible about “confidence” of finding sample mean within a certain precise interval
 - ▶ Improvement in bound will improve precision

Examples: Iris, Taj Mahal and IPL

- Iris data: Sepal length
 - ▶ $n = 50$, Sample mean: 5.006, Sample variance: 0.1242
 - ▶ With probability more than $1 - \sigma^2/50\delta^2$, sample mean lies in $[\mu - \delta, \mu + \delta]$
 - ★ Works for $\delta > \sigma/\sqrt{50}$

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$$1 - \frac{\sigma^2}{50\delta^2} > 0.95 \Rightarrow \delta^2 > \frac{\sigma^2}{50 \times 0.05}$$
$$\delta > \sqrt{\frac{1}{5}} \sigma$$

- Taj Mahal air quality: PM2.5

- ▶ $n = 11$, Sample mean = 65.72, Sample variance = 15.9²
- ▶ With probability more than $1 - \sigma^2/11\delta^2$, sample mean lies in $[\mu - \delta, \mu + \delta]$

- ★ Works for $\delta > \sigma/\sqrt{11}$

$$1 - \frac{\sigma^2}{11\delta^2} > 0.95 \Rightarrow \delta^2 > \frac{\sigma^2}{11 \times 0.05}$$
$$\delta^2 > \frac{20\sigma^2}{11}$$
$$\delta > \sqrt{\frac{20}{11}} \sigma$$

Examples: Iris, Taj Mahal and IPL

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 - ★ Works for $\delta > \sigma/\sqrt{11}$
- IPL: Runs scored in Delivery 0.3
 - ▶ $n = 1598$, Sample mean: 0.9524, Sample variance: 2.0666
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 - ★ Works for $\delta > \sigma/\sqrt{1598}$
- What to do when distribution is unknown? Have to assume something